

Reddit-Based Smartphone Brand and Model Analysis Report

Author(s): Arib Zayn Araf - 514550 & Anas Mahfooz - 514587

1. Introduction

This report details an in-depth analysis of public sentiment and discussion topics surrounding major smartphone brands and models (specifically Samsung Galaxy S-series and Apple iPhone series) on Reddit. Leveraging advanced Natural Language Processing (NLP) techniques, including LLM-based data labeling, DeBERTa for model identification, BERTopic for topic modeling, and DistilRoBERTa for sentiment analysis, we processed over 71,452 cleaned Reddit posts and comments. The project aimed to uncover prevailing sentiments, identify key discussion themes, and provide actionable insights for market research and product strategy. Our findings reveal distinct sentiment profiles and common user concerns across different models, highlighting areas for brand improvement and competitive advantages.

1.3 Goals and Objectives

Goal: To provide data-driven insights into consumer discussions and sentiments regarding the latest Samsung Galaxy S-series and Apple iPhone models on Reddit.

Objectives:

- To collect, clean, and preprocess relevant Reddit posts and comments related to specified smartphone brands and models.
- To develop and apply a machine learning model for accurate identification of specific phone models within unstructured text.
- To identify and categorize key discussion topics emerging from user conversations about these phones.
- To analyze the sentiment (positive, neutral, negative) associated with identified phone models and discussion topics.

2. Methodology

Our analytical pipeline involved several stages: data acquisition, comprehensive preprocessing, model identification, topic modeling, and sentiment analysis.

2.1 Data Acquisition and Preprocessing

Reddit data was programmatically collected using the PRAW library, targeting popular subreddits such as r/Android, r/Apple, r/GalaxyS23, r/GalaxyS24, r/iPhone, r/Smartphones, and r/samsung. Initial collection yielded over 6,616 posts and 75,473 comments.

Preprocessing Steps:

- **Duplicate Removal:** Identifying and eliminating redundant entries based on unique identifiers.
- **Temporal Filtering:** Retaining only content posted after June 1, 2023.
- **Content Filtering:** Removing content from deleted users, suspected bots, and entries not containing specific keywords related to the target phone brands.
- **Deep Text Cleaning:** Applying a series of transformations including lowercasing, URL removal, Reddit-specific markdown and noise removal (e.g., &,), non-alphabetic character stripping, and

normalization of whitespace.

- **Language Detection:** Filtering to ensure only English-language content was retained.

After preprocessing, a refined dataset of approximately 71,452 relevant posts and comments was available for further analysis.

2.2 Model Identification (DeBERTa Fine-tuning)

To accurately identify specific phone models (e.g., "iPhone 15 Pro Max", "Samsung S24 Ultra") within the text, we employed a fine-tuned transformer model.

- **Data Labeling:** To understand using context, which specific phone model a target comment or a target post is referring to, both directly and indirectly, we have used LLM Gemma 3 27B, powered by Google, which has an astounding capability to understand context.
- **Data Augmentation:** To address class imbalance, particularly for less frequently mentioned models, we applied data augmentation techniques.
- **Model Training:** We fine-tuned DeBERTa-base v3 for multi-label sequence classification. The model was trained on a stratified train-validation-test split of the labeled dataset. It was chosen instead of RoBERTa, which was our initial choice, simply because it was outperforming, reaching 0.69 F1 score, which is although not the best, was twice as much performance than RoBERTa. The performance could be better if we had used bigger dataset, but due to resource limitation, we used only a dataset of 1500 samples.

2.3 Topic Modeling (BERTopic)

To uncover the underlying themes and discussion points within the collected text data, we employed BERTopic. This technique leverages transformer embeddings and a class-based TF-IDF procedure to create dense clusters of documents, which are then represented by keywords. The resulting topics were manually reviewed and interpreted to assign coherent names (e.g., "Battery Life & Charging," "Software, UI & Notifications," "Camera Performance"). Irrelevant or overly general topics (e.g., "General Discussion," "Noise") were filtered out to focus on actionable insights.

2.4 Sentiment Analysis (DistilRoBERTa)

Sentiment analysis was performed using a pre-trained distilroberta-base-sentiment-classifier model. This model assigned a sentiment score (positive, neutral, or negative) to each cleaned Reddit post and comment. The raw sentiment outputs were then standardized into a consistent 'Positive', 'Neutral', and 'Negative' categorization for easier interpretation and aggregation.

3. Key Findings and Results

3.1 Model Identification Performance

The fine-tuned DeBERTa model demonstrated strong performance in identifying specific phone models from Reddit comments. On test set, the model achieved an average micro-F1 score of **0.681** and a macro-F1 score of **0.696**. Precision and recall metrics for individual classes were also high.

3.2 Dominant Discussion Topics

Our BERTopic analysis revealed several prominent discussion topics across the collected Reddit data. The top 3 most frequently discussed topics included:

1. **Flagship Phone Comparisons & Features:** Discussions centered around comparisons between

different flagship models, their overall features, and general phone-related discussions.

2. **Software Updates & User Interface:** Users frequently discussed operating system performance, user interface design, and notification management.
3. **Battery Performance & Charging:** Conversations revolved around battery longevity, charging speeds, and wireless charging capabilities.

3.3 Overall Sentiment by Brand

Overall sentiment analysis across the combined filtered dataset (comments and posts relevant to specific phone models) indicated a generally balanced tone, with a slight leaning towards negative and neutral discussions.

- **Overall Sentiment Distribution:**
 - Neutral: 42.27%
 - Negative: 34.83%
 - Positive: 22.89%

(A stacked bar chart comparing overall sentiment distribution for Apple vs. Samsung would be included here, if specific brand-level aggregated percentages were available in the analysis data.)

3.4 Sentiment Analysis by Model and Topic

We have plotted multiple bar chart graph, for each model in the x-axis and the percentage of sentiment on the Y-axis, for each topic, for in depth analysis, displaying how in each topic , the phone models influenced the brand perception, which shows us the weakness and strength of each models.

3.5 Sentiment over each brand over time by using models in chronological order

The graph gives us a clear idea of how people's perception on the brand changed over time after every new release of models, by checking the positive sentiment percentages for each models in chronological order. From this we can see that sentiment over Iphone is following a strict positive trend, with people's sentiment increasing significantly after each new models are introduced, while Samsung is fluctuating, giving us the idea that Samsung might not be consistent with their quality or marketing with every new models

4. Future Work

Things we missed out and would like to do to expand the project later on:

- Expanding data collection to other social media platforms (e.g., X/Twitter, forums) for a broader perspective.
- Implementing time-series analysis to track sentiment and topic trends over time, not just for assuming time for newer models showing more recent perception for brand.
- In-depth analysis for **Holder** by integrating user demographics for segmented analysis, understanding how perception of brand changes across different communities and region.